Hilbert space $\mathcal{H}$ which seperable

$\rightarrow$ $\mathcal{H}$ has a countable orthogonal basis $\{|i\rangle \mid i \in I\}$

- $\langle i|j\rangle = \begin{cases} 0 & \text{if } i \neq j \\ \neq 0 & \text{if } i = j \end{cases}$
- if $|\psi\rangle \in \mathcal{H}$ and $\langle \psi|i\rangle = 0 \ \forall i \in I$ then $|\psi\rangle = 0$

$\rightarrow$ $|\psi\rangle = \sum_i \frac{\langle i|\psi\rangle}{\langle i|i\rangle} |i\rangle$ $\rightarrow$ $\sum_i \frac{|i\rangle\langle i|}{\langle i|i\rangle} = \mathbb{1}$ projection operator on basis

$$\langle \psi|\psi\rangle = \sum_i \frac{\langle \psi|i\rangle\langle i|\psi\rangle}{\langle i|i\rangle} = \sum_i \frac{|\langle \psi|i\rangle|^2}{\langle i|i\rangle}$$

$$1 = \sum_i \frac{|\langle \psi|i\rangle|^2}{\langle \psi|\psi\rangle\langle i|i\rangle}$$

interpret $P(i) = \frac{|\langle \psi|i\rangle|^2}{\langle \psi|\psi\rangle\langle i|i\rangle}$

probability that $|\psi\rangle$ is measured in state $|i\rangle$

$$\sum_i P(i) = 1$$

$|\psi\rangle \sim \lambda|\psi\rangle$ if $\lambda \in \mathbb{C}^*$

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if $|\psi\rangle$ $|\psi'\rangle$, what is probability that $|\psi\rangle$ is measured as $|\psi'\rangle$?

$$P(\psi \rightarrow \psi') = \frac{|\langle \psi'|\psi\rangle|^2}{\langle \psi|\psi\rangle\langle \psi'|\psi'\rangle}$$ transition amplitude

. . . . . .

ex $T = 3\mathbb{1}$ linear and prop. to unitary

$P(T\psi \rightarrow T\psi') = P(\psi \rightarrow \psi')$

T does not preserve inner products: $\langle T\psi|T\psi'\rangle = 9\langle \psi|\psi'\rangle$

ergodic theorem $\quad U$ unitary operator on $\mathcal{H}$, $\psi \in \mathcal{H}$

average of $\{\psi, U\psi, U^2\psi, \dots\}$

$$\exists \langle\psi\rangle \in \mathcal{H} : \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} U^k \psi = \langle\psi\rangle$$

$\boxed{H \text{ Hamiltonian}}$ operator $\quad U_t = e^{iHt/\hbar}$ time evolution operator

$$|\psi(t)\rangle = U_t |\psi(0)\rangle$$

$$\frac{d}{dt}|\psi(t)\rangle = \left(\frac{d}{dt}(U_t)\right)|\psi(0)\rangle = \frac{i}{\hbar} H U_t |\psi(0)\rangle = \frac{i}{\hbar} H |\psi(t)\rangle$$

$$-i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle \quad \text{shrodinger}$$

Statistical physics: $\quad$ time average $=$ ensemble average

ergodic assumption

(over infinite time)

$\mathcal{H}^U$ = set U-invariant vectors $= \{|\psi\rangle \in \mathcal{H} \mid U|\psi\rangle = |\psi\rangle\}$

H is conserved trivially

only physical thing with A that we can measure

A operator is conserved if $\forall |\psi\rangle$ $\dfrac{\langle\psi(t)|A|\psi(t)\rangle}{\langle\psi(t)|\psi(t)\rangle}$ independent of t

$A : \mathcal{H} \to \mathcal{H}$ Hermitian

finite-dimensional version

spectral theorem: if $A : \mathcal{H} \to \mathcal{H}$ Hermitian then there is an orthonormal basis of $\mathcal{H}$ composed of eigenvectors of A: $\mathcal{B} = \{|i\rangle \mid i \in I\}$

such that $A|i\rangle = A_i|i\rangle$

average of A in state $|\psi\rangle$?

$$P(i) = \frac{|\langle\psi|i\rangle|^2}{\langle\psi|\psi\rangle} \quad \text{probability of measuring } |\psi\rangle \text{ in state } |i\rangle$$

↳ assumption of QM

- A_i (eigenvalue in state $|i\rangle$) is the value of A for state $|i\rangle$

$$\langle A\rangle = \sum_i A_i P(i) = \sum_i A_i \frac{\langle\psi|i\rangle\langle i|\psi\rangle}{\langle\psi|\psi\rangle} = \sum_i \frac{\langle\psi|A|i\rangle\langle i|\psi\rangle}{\langle\psi|\psi\rangle}$$

$$= \frac{\langle\psi|A|\psi\rangle}{\langle\psi|\psi\rangle}$$

$A : \mathcal{H} \to \mathcal{H}$ <u>conserved</u> if $\langle A\rangle_{\psi(t)} = \langle A\rangle_{\psi(0)} \quad \forall t, \forall \psi$

$$\langle A\rangle_{\psi(t)} = \frac{\langle\psi(t)|A|\psi(t)\rangle}{\langle\psi(t)|\psi(t)\rangle} = \frac{\langle\psi(0)|U_t^\dagger A U_t|\psi(0)\rangle}{\langle\psi(0)|\underbrace{U_t^\dagger U_t}_{=\mathbb{1}}|\psi(0)\rangle}$$

$$= \frac{\langle\psi(0)|U_t^\dagger A U_t|\psi_0\rangle}{\langle\psi(0)|\psi(0)\rangle}$$

$$= \frac{\langle\psi(0)|A|\psi_0\rangle}{\langle\psi_0|\psi_0\rangle} \quad \forall \psi_0, \forall t$$

$$U_t^\dagger A U_t = A \quad \forall t \quad \Rightarrow \quad e^{-iHt/\hbar} A e^{iHt/\hbar} = A$$

$\left.\frac{d}{dt}\right|_{t=0}$ on both sides : $-\frac{iHA}{\hbar} + \frac{iAH}{\hbar} = 0 \iff [A, H] = 0$

<u>Noether's theorem</u> : if you write QM into Lagrangian theory, we have a symmetry: <u>multiplication by a phase</u> → conserved quantity $\|\psi\rangle\|$

↳ symmetry of theory

↳ if we have a Lagrangian $\mathcal{L}$ describing a theory and an infinitesimal symmetry of $\mathcal{L}$ (or more generally of the action $\int \mathcal{L}\, dt$) then we have a conserved quantity

$U_t = e^{itH/\hbar}$ invariant vectors $|\psi\rangle$ satisfy $U_t|\psi\rangle = |\psi\rangle$

let $\{|i\rangle \mid i \in I\}$ be a basis of eigenvectors of H

$H|i\rangle = E_i|i\rangle$ → energy of state $|i\rangle$

$|\psi\rangle = \sum_i \boxed{\langle i|\psi\rangle}|i\rangle$

$U_t|i\rangle = e^{-itH/\hbar}|i\rangle = (\mathbb{1} - itH/\hbar - t^2H^2/2\hbar + \dots)$

$= e^{-itE_i/\hbar}|i\rangle$

$U_t|\psi\rangle = \sum_i \langle i|\psi\rangle U_t|i\rangle$

$= \sum_i \langle i|\psi\rangle e^{-itE_i/\hbar}|i\rangle$

$\langle i|\psi\rangle = \langle i|\psi\rangle e^{-itE_i/\hbar} \quad \forall i$ with $t>0$ and $\{E_i \mid i \in I\}$ fixed

$\langle i|\psi\rangle = 0$ if $E_i \neq 0$

$|\psi\rangle$ is an eigenstate of H with eigenvalue zero

$|\psi\rangle \in \ker(H)$

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suppose $H|\psi\rangle = 0$ then $e^{-iHt/\hbar}|\psi\rangle = |\psi\rangle$

$\Rightarrow$ in MET the projection on $\mathcal{H}^U$ is projection on kernel of H

<u>Harmonic oscillator</u> $H = \omega\left(a^+a + \frac{1}{2}\mathbb{1}\right)$

basis $\{|n\rangle \mid n \in \mathbb{N}_0\}$ $H|n\rangle = \omega(n+\frac{1}{2})|n\rangle$

$U_t = e^{-itH}$ $\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} e^{-itkH}|\psi\rangle = 0$ $\Delta t = \frac{1}{n}$

$t \mapsto e^{-itH}$

$\rightsquigarrow \lim_{T\to\infty} \frac{1}{T}\int_0^T \underbrace{e^{-itH}|\psi\rangle}_{|\psi(t)\rangle} dt \in \{|\varphi\rangle \in \mathcal{H} \mid U_t|\varphi\rangle = |\varphi\rangle \ \forall t\}$

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-itH}|\psi\rangle\, dt = |\psi_0\rangle$$

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-it(H+\lambda\mathbb{1})}|\psi\rangle\, dt$$

1-dim. Hilbert space: $\mathcal{H} = \mathbb{C}$ $\qquad H: z\in\mathbb{C} \mapsto z z \in \mathbb{C}$

$U_t: z\in\mathbb{C} \mapsto e^{-izt} z \in \mathbb{C}$

$$\lim_{T\to\infty} z \frac{1}{T}\int_0^T e^{-izt}\, dt = 0$$

$H: z\in\mathbb{C} \to 0 \in \mathbb{C}$ $\qquad \ker H = \mathbb{C}$

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T z\, dt = z$$

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action of U_t on A (adjoint action)

$$\left[H, \lim_{T\to\infty} \frac{1}{T}\int_0^T dt\, U_t^\dagger A U_t\right] \overset{?}{=} 0$$

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T d\tau\, [H, U_t^\dagger A U_t] = 0\ ?$$